Enrollment, Tuition Subsidies and Welfare: Evidence from Ontario's Tuition Grant

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Research Question

- Do tuition subsidies increase enrollment in Post Secondary Education (PSE) institutions?
- If so, what are the mechanisms through which tuition subsidies increase enrollment?

Motivation

- We know little about the mechanisms through which these subsidies work.
- Subsidies for PSE are designed to increase access but the literature predominantly focuses on the United States.

This Paper

1. Documents the effects of 30% Ontario Tuition Grant on enrollment in PSE
   - recipients enrolling in a 4-year university (resp. 2-year college) program receive $1,680 (resp. $780) each year.
   - my analysis focuses on 1.5 generation immigrants for whom I can link student education and income tax returns to the tax returns of their parents
2. Adopts DDD design and uses administrative tax-post-secondary data to compare the probability of enrolling in PSE program in Ontario for those who/whose:
   1. *parental income* is below the threshold $160,000 to those whose is above (first difference),
   2. *reside* in Ontario to those who do not (second difference),
   3. *enrolled* before and after the policy's introduction in 2012 (third difference)
3. My baseline estimates suggest that eligibility for the OTG increased PSE enrollment by 3pp
   - Students likely to be liquidity constrained respond more to the subsidy.
4. Sufficient statistics welfare analysis: evaluated from current levels, increasing tuition subsidies by $1 raises welfare by $1.32

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Contributions

1. Literature on the impact of tuition subsidies on enrollment and the role of liquidity constraints (Dynarski (2003), Dynarski and Scott-Clayton 2022, Page and Scott-Clayton 2015, Cornwell, Mustard & Sridhar, 2006, Cook, 2024)
   - Compared with previous work in the Canadian setting, better identification strategy due to richer data AND welfare analysis
   - Compared with previous papers using U.S. data, I focus on an understudied (and positively selected) sub-population (i.e. immigrants in Canada vs. USA)

2. Literature that studies age at immigration effects for landed immigrant children (second and 1.5 generation) (Connolly et. al., 2019, 2022,2023, Schaafsma, J., Sweetman 2001,) and children refugees (Zheng, A et. al. 2023)
   - Enrollment in PSE may be a channel through which immigrants integrate
   - New evidence on enrollment responses by landing age and world area of birth

Outline

1. Introduction
2. Institutional Background
3. Data
4. Research Design
5. Results
   - Main Results
   - Mechanisms
   - Robustness
6. Welfare Analysis
7. Conclusion

30% Ontario Tuition Discount

Implemented in January 2012 following a 2011 provincial election promise by the incumbent political party *(Cook, 2024)*

- eligibility criteria:
  1. enrolled full-time at a public college or university in Ontario, in a first entry program,
  2. were a Canadian citizen/permanent resident/protected person,
  3. Ontario resident with parents,
  4. had left high school within the last four years,
  5. gross parental income was $160,000 or less,
- five out of every six students in Ontario would be eligible,

Data

Two administrative datasets from the ELMLP (Education and Labour Market Longitudinal Platform)

1. IMDB-T1ff (2009-2014)
   - links administrative immigration records with individual tax returns of 1.5 generation immigrants and their immigrant parents
   - contains information on income, family demographics
2. Post Secondary Information System (2009-2014)
   - records detailed information on enrollments and graduates of Canadian public post-secondary institutions
   - individual level enrollments, program description

IMDB-PSIS linkage: pooled cross-sections of first-time post-secondary entrants (aged 18-22) for each academic year from 2009–2014

- dependent variable: ever enrolled in a degree program in Ontario PSE institute.

Mean Differences: Ontario | Mean Differences: ROC | Institutes in PSIS | Immigrants in PSIS vs IMDB | 1.5 Gen Immigrants by Institution Type

Research Design

$$
\begin{aligned}
Y_{ipt} = \alpha &+ \beta_1 \, \text{TREAT}_i + \beta_2 \, \text{POST}_t + \beta_3 \, \text{REZ}_{ipt} + \beta_4 \, (\text{TREAT}_i \times \text{POST}_t) \\
&+ \beta_5 \, (\text{TREAT}_i \times \text{REZ}_{ipt}) + \beta_6 \, (\text{POST}_t \times \text{REZ}_{ipt}) \\
&+ \beta_7 \, (\text{TREAT}_i \times \text{POST}_t \times \text{REZ}_{ipt}) \\
&+ \gamma_p + \delta_t + \varepsilon_{ipt}.
\end{aligned} \tag{1}
$$

where,

- Y_{ipt}: 1 if child i in province p at time t enrolls in a degree program in Ontario PSE
- REZ_{ipt}: 1 if child i resides in Ontario with main parent residing in Ontario at time t and 0 otherwise
- $TREAT_i$: 1 if gross parent income of child $\leq$ \$160000
- $POST_t$: 1 if year $t \geq 2012$

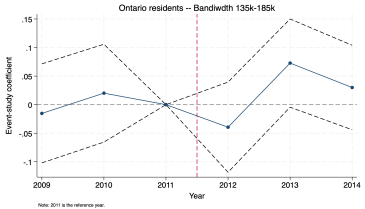
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\end{aligned}
\tag{2}
$$

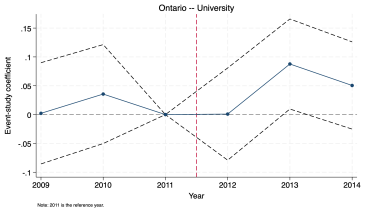
- identifying assumption: absent the policy, the difference in enrollment trends between income-eligible and income-ineligible immigrants would have evolved similarly in Ontario and in the rest of Canada
- restrict the estimation sample to a narrow income bandwidth around the eligibility cutoff ($135,000 -$185,000)
  1. children on either side of the cutoff come from families that are observationally very similar
  2. policy-relevant margin, as for students for whom eligibility status is most salient and for whom the grant is most likely to shift the enrollment decision

Event study: Ontario | Event study: ROC below 160,000

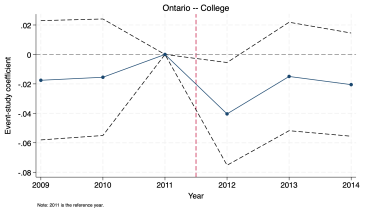
Event Study: Ontario ($135,000-$185,000)



(a) Enrollment



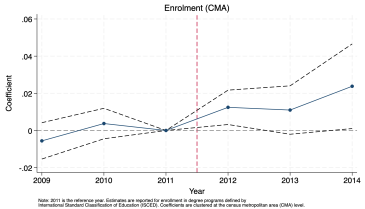
(b) Enrollment in University



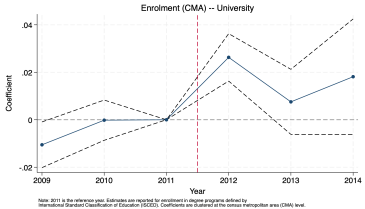
(c) Enrollment in College

Return

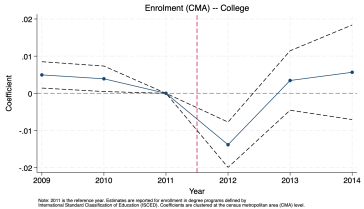
Event Study: Below Policy Threshold (160,000)



(a) Enrollment



(b) Enrollment in University



(c) Enrollment in College

Return

Enrollment Response to Subsidy by Institute Type

	Any Ontario PSE			Ontario University			Ontario College		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treatment × Post × Ontario	0.027	0.030	0.032	0.028	0.030	0.040	-0.001	-0.002	-0.008
	(0.005)	(0.005)	(0.010)	(0.005)	(0.005)	(0.005)	(0.001)	(0.001)	(0.002)
	[0.084]	[0.069]	[0.040]	[0.100]	[0.082]	[0.073]	[0.610]	[0.278]	[0.165]
Child Controls		X	X		X	X		X	X
Parent Controls			X			X			X
Observations	15,960	15,220	14,770	15,960	15,220	14,770	15,960	15,220	14,770
Enrollment rate	0.370	0.370	0.370	0.340	0.340	0.340	0.030	0.030	0.030

DiD ON | DiD ROC | DiD Siblings | STEM-BHASE | Migration ON | MVPF | Intensive Margin

Mechanism: Why Does the Subsidy Increase Enrollment?

- What mechanisms can explain the positive enrollment response to the Ontario tuition subsidy?
  1. one direct mechanism is that it relaxes liquidity and borrowing constraints faced by students from lower-asset households,
  2. another mechanism is that the subsidy indirectly helps solve an information problem, *(Oreopoulos & Dunn, 2013)*

- To probe these mechanisms, I examine whether enrollment responses are larger among,
  1. students whose parents report less capital income during ages 15–17, a proxy facing liquidity constraints,
  2. students reporting RESP withdrawls/income OR scholarship income in their tax filings at age 18, a proxy for being less likely to face liquidity constraints

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The Role of Liquidity Constraints

	Parent Capital Income			Proxy for RESP/ Scholarship Income	
	Bottom Tercile	Middle Tercile	Upper Tercile	Liquidity Constrained	Not Liquidity Constrained
Treatment × Post × Ontario	0.032 (0.010) [0.095]	0.010 (0.014) [0.929]	0.018 (0.012) [0.431]	0.047 (0.009) [0.082]	-0.043 (0.011) [0.094]
Child Controls	X	X	X	X	X
Parent Controls	X	X	X	X	X
Observations	6,540	1,020	7,110	8,610	2,315
Enrollment rate	0.348	0.388	0.395	0.357	0.253

Event Study: Ontario by Capital Income | Event Study: ROC by Capital Income | Capital Income (IHS)

Robustness and Sensitivity Analyses

1. Bandwidth Sensitivity around the Eligibility Cutoff
   - alternative bandwidths: ±$35k and ±$45k around the $160k threshold
   - estimates remain stable for moderate expansions; significance attenuates with wider bandwidths, consistent with reduced local comparability ⏩ RESULTS
2. Placebo threshold windows
   - test sensitivity to placebo income cutoffs ($50k, $60k, . . . , $100k)
   - evaluates whether results are driven by observations close to the policy threshold ⏩ RESULTS
3. Placebo province tests
   - assign placebo treatment status to non-treated provinces (e.g., British Columbia and Alberta)
   - no significant enrollment response in placebo provinces supports the Ontario-specific policy interpretation ⏩ RESULTS
4. Manipulation / Density test at the threshold
   - implement a local polynomial density discontinuity test *(Cattaneo, Jansson & Ma 2020)*
   - tests for strategic income reporting or sorting around eligibility ⏩ RESULTS

Heterogeneity

- Next, I examine heterogeneity in subsidy response by 1.5 generation immigrant children by their
  1. World area of birth.
     - on average, a child immigrated from Africa & Middle-East is 15.7 pp more likely to respond to subsidy for PSE in Ontario post 2012 ⏩ RESULTS
  2. Landing age
     - children arriving earlier have a higher response to the subsidy but not statistically significant ⏩ RESULTS

Welfare Analysis

This paper adopts a two-period life-cycle model following Lawson (2017). Some high-level details of the model:

- Partial equilibrium (wages fixed and production not modelled)
- In period 1, a young person decides whether to attend college or not ($s_i \in \{0, 1\}$). Students differ in their parental resources (A_i)
- In period 2, they choose labor supply l
- Student enrolls if $Utility(s = 1) > Utility(s = 0)$
- Government chooses a tuition subsidy b and a proportional labor income tax rate τ

The objective is to quantify the relative channels of the mechanisms through which a tuition subsidy may affect welfare

Details

Welfare Gain Formula

I consider the effect of a small increase to the tuition subsidy b, starting from its current level $\bar{b}$

$$\left.\frac{dW}{db}\right|_{b=\bar{b}} \simeq S \times \left[\underbrace{L}_{\text{Liquidity Effect}} \underbrace{- \left(\frac{b+p}{b}\right) \epsilon_{Sb} + \frac{\tau S}{b} \epsilon_{Sb} \Delta Y_s}_{\text{Net Fiscal Externality}} \right] \tag{3}$$

- Liquidity effect: **L** measures the welfare effect of transferring \$1 to individuals from period 1 to period 0

$$L = \frac{E_1[u'(c_{ui})] - v^0(c_v^0, l_0)}{v^0(c_v^0, l_0)} = \frac{\frac{\partial S}{\partial A}}{\frac{\partial S}{\partial b} - \frac{\partial S}{\partial A}} \tag{4}$$

- Net Fiscal Externality:

Welfare Gain Formula

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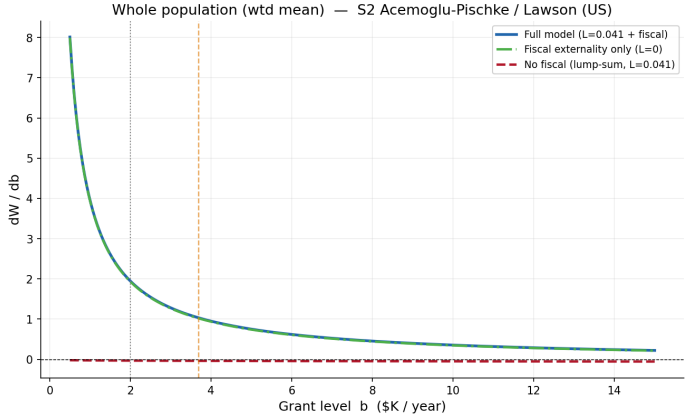
- Liquidity effect: **L** measures the welfare effect of transferring \$1 to individuals from period 1 to period 0
- Net Fiscal Externality:
  - additional tuition subsidy (b) plus other costs (p) due to marginal students enrolling in PSE multiplied by the elasticity of enrollment w.r.t. the subsidy (i.e. $\epsilon_{Sb} = \frac{dS}{db} \times \frac{b}{S}$)
  - additional tax revenue associated with the earnings premium from PSE (ΔY_s)

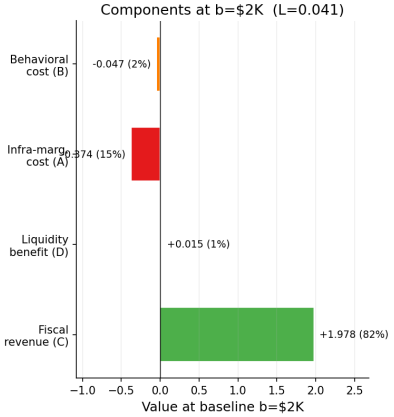
Takeaway: ϵ_{Sb} relevant for welfare

Baseline Calibration Parameters

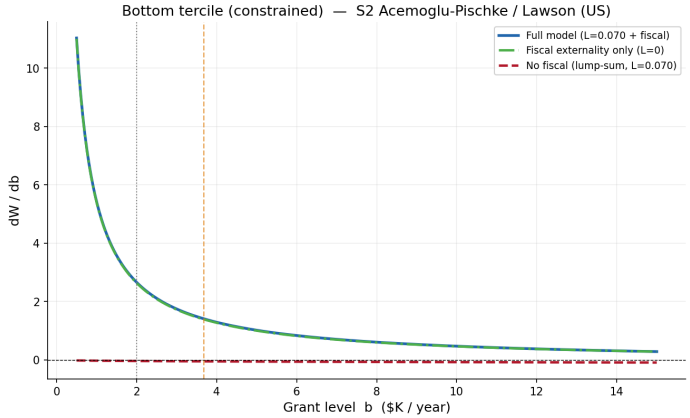
Parameter	Description	Value
S	Baseline enrollment rate	0.37
$\hat{\beta}^{DDD}$	Triple-difference estimate (bottom tercile)	0.032
Δb	Additional grant induced by policy ($1,000s)	1.68
b	Baseline annual grant ($1,000s)	2.00
e	Annual tuition cost ($1,000s)	7.24
$\frac{\partial S}{\partial a_1}$	Effect of income on enrollment	$[0, 0.0021]$
τ	Effective tax rate	0.28
r	Annual discount rate	0.03
ΔY_s	Present value of lifetime earnings gain ($1,000s)	90.00

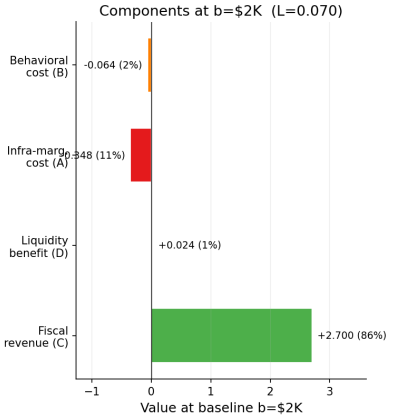
Decomposition of Welfare Gain



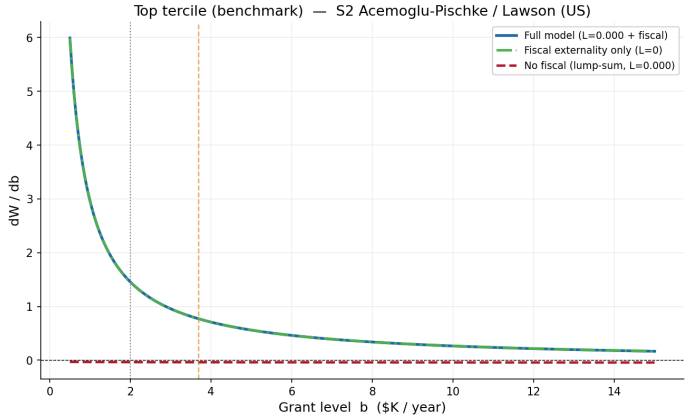


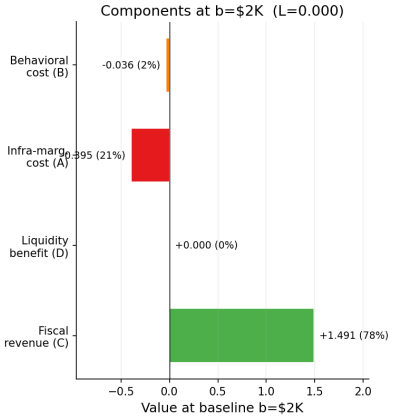
Decomposition of Welfare Gain: Bottom Tercile





Decomposition of Welfare Gain: Upper Tercile





Conclusion

- DDD estimates are indicative that the 30% OTG increased enrollment of 1.5 generation immigrants by 3 pp
  - with baseline enrollment at 37%, enrollment increased by 8.82% that corresponds to approximately 343 additional enrolled students in Ontario post-subsidy.
  - immigrant children that are relatively more financially constrained *(no capital income and RESP collections)* respond meaningfully than those that do not
- Welfare Analysis: My welfare calculations suggest that each additional dollar of grant spending raises social welfare by about $1.32
  - welfare decomposition reveals that fiscal externality is the dominant channel
  - the subsidy pays for itself

Enrollment Response to Subsidy by Institute & Program Type

	Any Ontario PSE		Ontario University		Ontario College	
	STEM	BHASE	STEM	BHASE	STEM	BHASE
Treatment × Post × Ontario	0.001	0.005	0.005	0.007	-0.005	-0.002
	(0.007)	(0.005)	(0.007)	(0.005)	(0.001)	(0.001)
	[0.976]	[0.641]	[0.684]	[0.834]	[0.255]	[0.151]
Child Controls	X	X	X	X	X	X
Parent Controls	X	X	X	X	X	X
Observations	14,770	14,770	14,770	14,770	14,770	14,770
Enrollment rate	0.370	0.370	0.370	0.370	14,770	14,770

Return

Enrollment Response – Ontario ($135,000-$185,000)

	Enrollment response		Liquidity Constrained		Not Liquidity Constrained	
	CMA	PARENT IMDB-ID	CMA	PARENT IMDB-ID	CMA	PARENT IMDB-ID
TREAT × POST	0.023*	0.022	0.004	0.006	0.033**	0.032
	(0.013)	(0.023)	(0.020)	(0.040)	(0.014)	(0.031)
Child Controls	X	X	X	X	X	X
Parent Controls	X	X	X	X	X	X
Observations	7,016	6,980	2,990	2,970	3,790	3,770

Return

Enrollment Response – Below Policy Threshold ($160,000)

	ENROLLMENT	UNIVERSITY	COLLEGE	LIQUIDITY CONSTRAINED	NOT LIQUIDITY CONSTRAINED
REZ × POST	0.038*	0.040*	-0.001	0.054**	0.034*
	(0.007)	(0.005)	(0.007)	(0.005)	(0.001)
	[0.053]	[0.056]	[0.946]	[0.049]	[0.038]
Child Controls	X	X	X	X	X
Parent Controls	X	X	X	X	X
Observations	177,085	177,085	177,085	104,470	72,620
Enrollment rate	0.317	0.277	0.039	0.293	0.351

Return

Enrollment Response – Below Policy Threshold ($160,000) – Siblings

	Enrollment response		Liquidity Constrained		Not Liquidity Constrained	
	CMA	PARENT IMDB-ID	CMA	PARENT IMDB-ID	CMA	PARENT IMDB-ID
REZ × POST	0.020	0.020***	0.019	0.019***	0.006	0.006
	(0.014)	(0.003)	(0.013)	(0.005)	(0.006)	(0.008)
Child Controls	X	X	X	X	X	X
Parent Controls	X	X	X	X	X	X
Observations	56,425	57,990	30,030	30,990	16,110	16,430
Enrollment rate	0.296	0.295	0.276	0.275	0.336	0.335

Return

Heterogeneous Effects by Landing Age

	0–11	12–19
Treatment × Post × Ontario	0.031	-0.003
	(0.015)	(0.022)
	[0.130]	[0.842]
Child controls	X	X
Parent controls	X	X
Observations	12660	2580
Enrollment rate	0.415	0.160

Standard errors in parentheses, clustered at the provincial level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

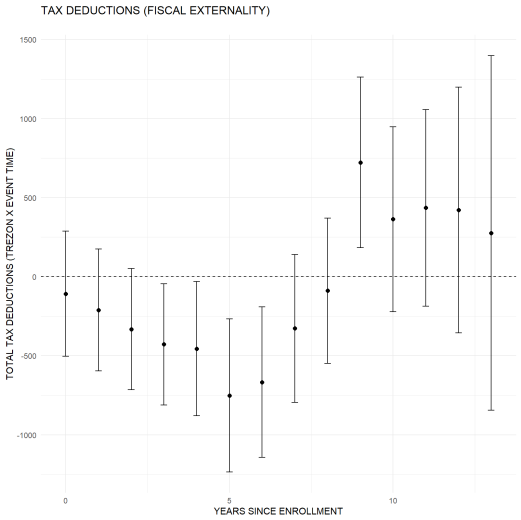
Return

Heterogeneity I - World Area of Birth

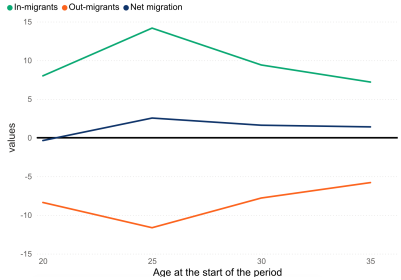
	Heterogeneity by World Area of Birth					
	Africa-Middle East	East Asia	Europe	Oceania-other Asia	South-Central America	South Asia
Treatment × Post × Ontario	0.157	-0.037	-0.055	0.067	0.138	-0.067
	(0.031)	(0.027)	(0.047)	(0.027)	(0.064)	(0.043)
	[0.043]	[0.290]	[0.611]	[0.200]	[0.182]	[0.311]
Child Controls	X	X	X	X	X	X
Parent Controls	X	X	X	X	X	X
Observations	2845	2270	3545	2730	1015	2260
Enrollment rate	0.407	0.462	0.324	0.223	0.354	0.515

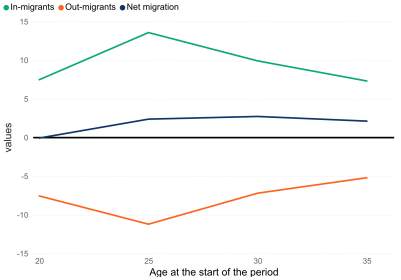
Return

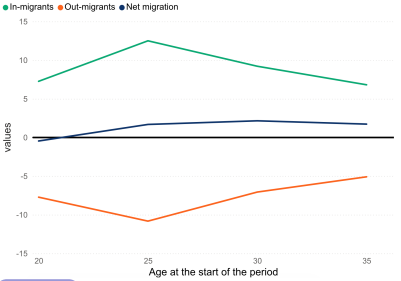
Larger Tax Base

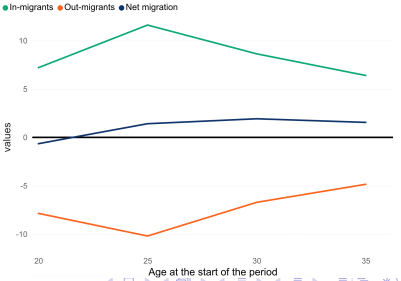


Inter-provincial Migration In/Out of Ontario









Return

MVPF

$MVPF = \frac{WTP}{NetCost}$	Description	Value
(A)	Program (mechanical) cost	$1680
(B)	Enrollment crowd-in cost	273.6
(C)	Tax FE (MTE: 28%)	703.2
	Net cost (A + B - C)	$1250.4
(D)	WTP (no internality)	$1680
(E)	Liquidity internality	390.6
	WTP (with internality: D + E)	2,070.6

Liquidity internality: A dollar delivered to a constrained household is worth more than a dollar, because their marginal utility of cash is elevated. That's the internality term in my WTP, and it's what lifts the MVPF from 1.3 to 1.7.

Return

MVPF

δ	α	internality=$\delta \times \alpha \times 0.62 \times 0.37$	$\frac{dW}{db} = 0.37 + L$	$MVPF = \frac{\frac{dW}{db}}{0.478}$
0	0.50	0	0.37	0.77
0	0.75	0	0.37	0.77
0	1.00	0	0.37	0.77
0.5	0.50	0.057	0.42	0.89
0.5	0.75	0.086	0.08	0.95
0.5	1.00	0.114	0.11	1.01
1	0.50	0.114	0.11	1.01
1	0.75	0.172	0.17	1.13
1	1.00	0.229	0.22	1.25

$costbaseFE/\Delta Y_S = 0.52\frac{3780}{90000} = 0.478$. δ is the weight society places on liquidity internality whereas α is how much a constrained student values the \$1 from subsidy.

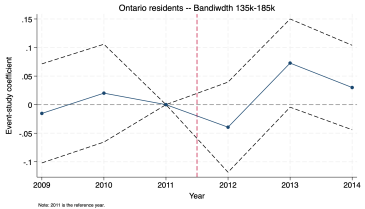
Return

Enrollment Response to Subsidy – Intensive Margin

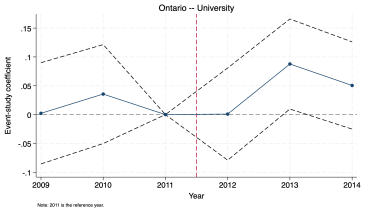
	ONTARIO ($135,000-$185,000)		ONTARIO + ROC (Below $160,000)		SIBLINGS (Below $160,000)	
	Constrained	Not Constrained	Constrained	Not Constrained	Constrained	Not Constrained
DiD	0.011	0.012	-0.004	-0.027	0.021	0.000
	(0.089)	(0.046)	(0.015)	(0.011)	(0.017)	(0.007)
			[0.501]	[0.885]		
Child Controls	X	X	X	X	X	X
Parent Controls	X	X	X	X	X	X
Observations	3,750	3,740	41,455	36,930	7,180	5,770

Return

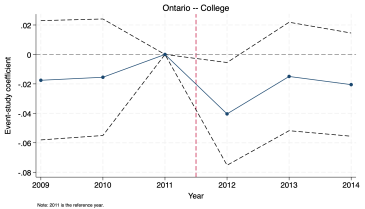
Event Study: Ontario ($135,000-$185,000)



(a) Enrollment



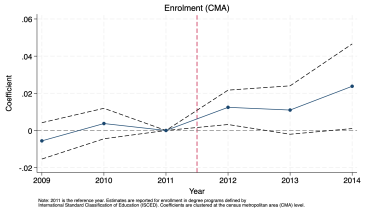
(b) Enrollment in University



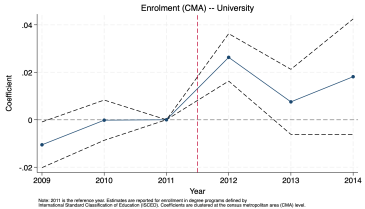
(c) Enrollment in College

Return

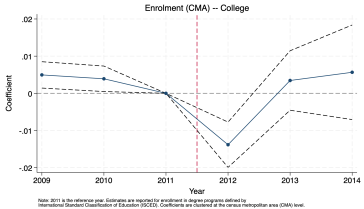
Event Study: Below Policy Threshold (160,000)



(a) Enrollment



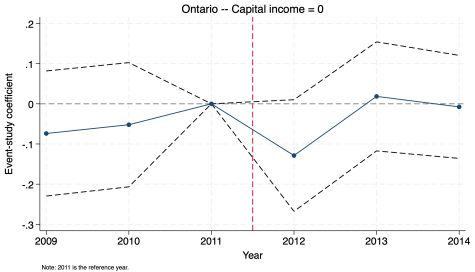
(b) Enrollment in University



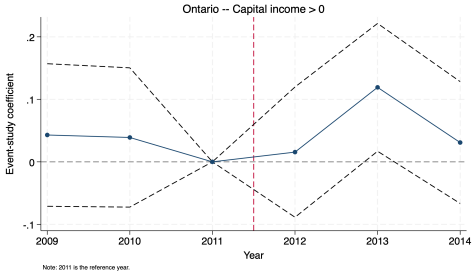
(c) Enrollment in College

Return

Event Study: Ontario ($135,000-$185,0000) by Capital Income



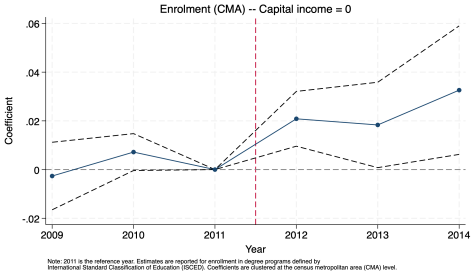
(a) Liquidity Constrained



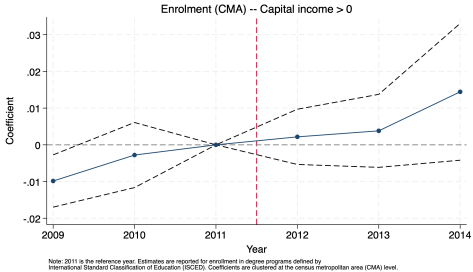
(b) Not Liquidity Constrained

Return

Event Study: Below Policy Threshold ($160,000)



(a) Liquidity Constrained



(b) Not Liquidity Constrained

Return

Mean Differences: Ontario residents ($135,000-$185,000)

	Diff. (Inelig.–Elig.)	SE
Panel A. Student-Parent characteristics		
Landing Age	-0.308***	(0.081)
Mother tongue: English	0.006	(0.008)
Mother tongue: Non-English	-0.006	(0.008)
Years of Schooling (child)	-0.117**	(0.049)
Years of Schooling (parent)	0.321***	(0.070)
Panel B. Student-Parent Incomes and Deductions		
Gross Parent Income (t-1)	24779.680***	(154.452)
Capital Income (parent)	92.988**	(43.038)
Education Deductions (parent)	35.800	(76.112)
Tuition Deductions (child)	269.649***	(73.753)
RESP-Scholarship Income (child)	27.509	(21.728)
Panel C. Enrollment		
Enrolled in PSE	0.014	(0.010)
Enrolled in ON PSE	0.026**	(0.011)
Enrolled in ON UNI	0.028**	(0.011)
Enrolled in ON COL	-0.002	(0.005)
Enrolled in ON STEM	0.021**	(0.010)
Enrolled in ON Non-STEM	0.010	(0.007)
N	9300	

Mean differences with standard errors in parentheses, * p <0.1, ** p<0.05 *** p<0.01

Return

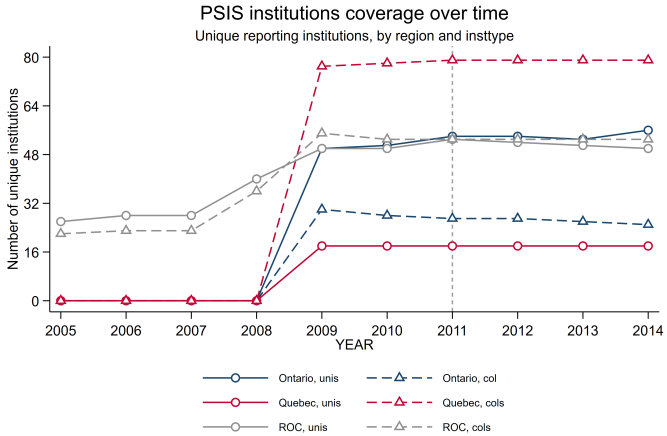
Mean Differences: Below Policy Threshold ($160,000)

	Diff. (ROC–ON)	SE
Panel A. Student-Parent characteristics		
Landing Age	0.662***	(0.018)
Mother tongue: English	0.008***	(0.001)
Mother tongue: Non-English	-0.008***	(0.001)
Years of Schooling (child)	0.408***	(0.013)
Years of Schooling (parent)	-0.299***	(0.017)
Panel B. Student-Parent Incomes and Deductions		
Gross Parent Income (t-1)	-5151.922***	(159.780)
Capital Income (parent)	231.683***	(14.296)
Education Deductions (parent)	-1123.599***	(11.046)
Tuition Deductions (child)	-656.310***	(12.664)
RESP-Scholarship Income (child)	14.671***	(3.592)
Panel C. Enrollment		
Enrolled in PSE	-0.289***	(0.002)
Enrolled in ON PSE	-0.427***	(0.002)
Enrolled in ON UNI	-0.362***	(0.001)
Enrolled in ON COL	-0.065***	(0.001)
Enrolled in ON STEM	-0.187***	(0.001)
Enrolled in ON Non-STEM	-0.092***	(0.001)
N	259360	

Mean differences with standard errors in parantheses, * p<0.10, ** p<0.05 *** p<0.01

Return

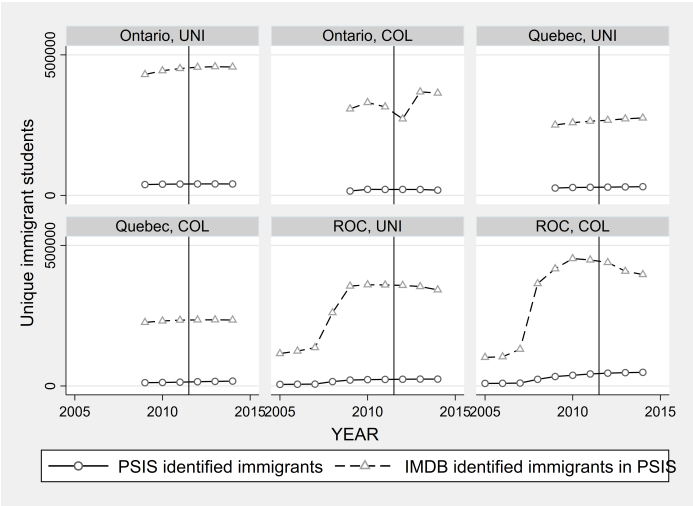
PSIS Coverage of Institutes Over Time



Note: PSIS national/provincial coverage around {}2009; univerisities (observed SEP-DEC) post 2008

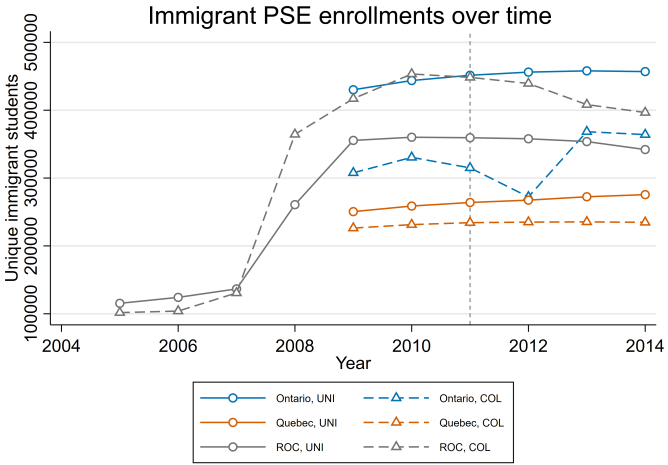
Return

PSIS vs. IMDB+PSIS Coverage of 1.5 Gen Immigrants



Return

PSIS Coverage of 1.5 Gen Immigrants Over Time



Return

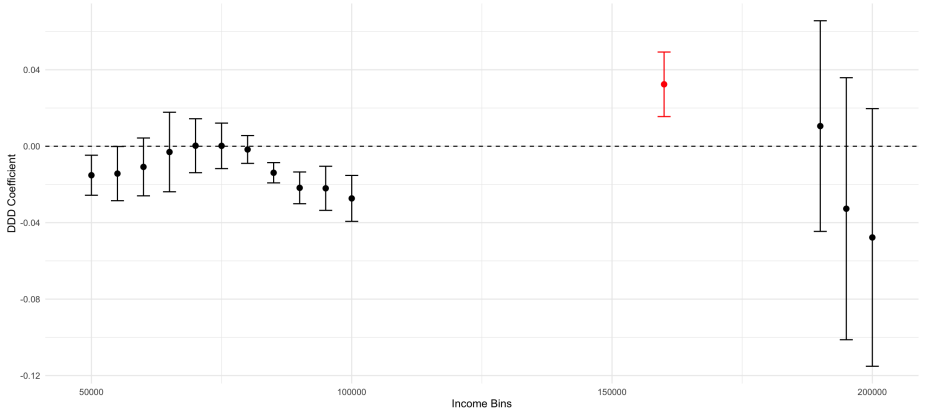
Robustness - I

	Income Bands ($000s)			
	135–185	125–195	115–205	105–215
Treatment × Post × Ontario	0.032 (0.010) [0.040]	0.021 (0.006) [0.053]	0.014 (0.007) [0.188]	0.004 (0.003) [0.186]
Child Controls	X	X	X	X
Parent Controls	X	X	X	X
Observations	14.770	22,115	29,895	38,955
Enrollment rate	0.340	0.369	0.368	0.324

Notes: Each column reports estimates from the baseline triple-difference specification using alternative parental income bandwidths around the tuition subsidy eligibility cutoff. All specifications include child, parent, and immigrant demographic controls. Significance levels are based on wild cluster bootstrap inference using the Webb six-point distribution with clustering at the provincial level. Conventional standard errors are reported in parentheses and Wild bootstrap p values in square brackets.

Return

Robustness - II - Placebo Income Bins



Return

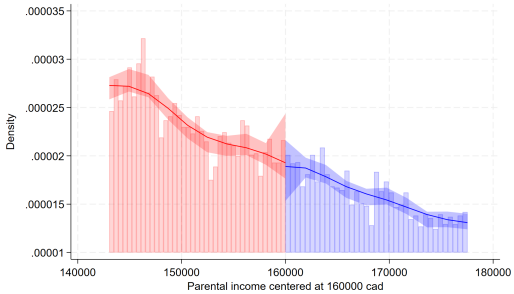
Robustness III - Placebo Provinces

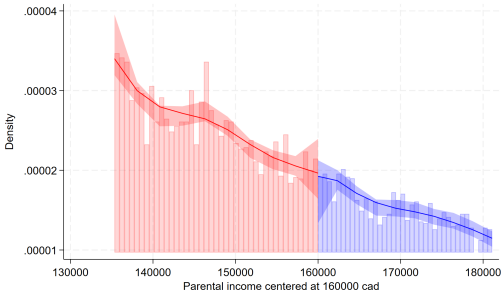
	Alberta	British Columbia
Treatment × Post × Ontario	0.008	0.004
	(0.006)	(0.003)
Child Controls	X	X
Parent Controls	X	X
Observations	14,770	14,770

Notes: Each column reports placebo estimates using Alberta and British Columbia as alternative comparison provinces. All specifications include child and parent demographic controls. Significance levels are based on wild cluster bootstrap inference using the Webb six-point distribution with clustering at the provincial level. Conventional standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Return

Robustness - IV - Cattaneo, Jansson & Ma 2020





Return

Appendix: Heterogeneity (PSE type — STEM-BHASE) - Capital Income

	ENROLLMENT BY PSE TYPE AND STEM-BHASE CATEGORIES WITH CAPITAL INCOME ≥ 0							
	UNI	COL	STEM	STEM UNI	STEM COL	BHASE	BHASE UNI	BHASE COL
> 0	0.033**	-0.011**	-0.031**	-0.028**	-0.003**	0.027***	0.034***	-0.007*
	(0.015)	(0.004)	(0.012)	(0.013)	(0.001)	(0.007)	(0.006)	(0.003)
N	5955	5955	5955	5955	5955	5955	5955	5955
= 0	0.037***	-0.012**	0.012	0.018**	-0.006**	-0.013**	-0.007	-0.005***
	(0.006)	(0.004)	(0.008)	(0.008)	(0.002)	(0.005)	(0.005)	(0.001)
CHILD	X	X	X	X	X	X	X	X
PARENT	X	X	X	X	X	X	X	X
N	8480	8480	8480	8480	8480	8480	8480	8480

Standard errors in parentheses, clustered at the provincial level. Results are robust with parent, child and immigrant characteristics. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Return

Appendix: Capital Income Variables

	TAX VARIABLES OF PARENTS CAPITAL INCOME				
	CAPITAL GAINS LOSSES (PARENT)	CAPITAL GAINS LOSSES (FAMILY)	INTEREST INVESTMENT	DIVEDEND (PARENT)	DIVEDEND (FAMILY)
$	0.09	-0.00	0.21	0.06	-0.10
	(0.07)	(0.12)	(0.12)	(0.16)	(0.12)
RESPONSE	0.01	0.01	0.03	0.019	-0.01
	(0.01)	(0.01)	(0.02)	(0.02)	(0.01)
CHILD	X	X	X	X	X
PARENT	X	X	X	X	X
N	14770	14770	14770	14770	14770

Standard errors in parentheses, clustered at the provincial level. Results are robust with parent, child and immigrant characteristics. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Return

Lawson (2017) Model

- individual's maximization problem

$$\max_{s_i \in \{0,1\},\, c^0,\, c_1^u,\, c^u,\, l_0,\, l_1} V_i = s_i \Big[u(c_{ui}) + R_2 v^1(c_{vi}^1, l_{1i}) + \eta_i - \lambda_{1i}(c_{ui} + R_2 c_{vi}^1 - (b-e) - (1-\tau)\gamma_2 w_{11} l_{1i}) - \mu_i(c_{ui} - (b-e) - A_i) \Big] + (1-s_i) \Big[R_1 v^0(c_{vi}^0, l_{0i}) - \lambda_{0i}(R_1 c_{vi}^0 - (1-\tau)\gamma_1 w_{01} l_{0i}) \Big] \tag{6}$$

- the government chooses b and τ subject to BC:

$$S(b+p) + G = \tau[S\gamma_2 E(Y_{11i}|s_i = 1) + (1-S)\gamma_1 E(Y_{01i}|s_i = 0)] = \tau Y \tag{7}$$

Return

Lawson (2017) Model

- FOCs:

$$\frac{\delta V_i}{\delta s_i} := [u(c_{ui}) + R_2 v^2(c^1_{vi}, l_{1i} + \eta_i - \lambda_{1i}(c_{ui}....)]] = 0$$
$$U_{1i} = U_{0i} \tag{8}$$

$$\frac{\delta V_i}{\delta c^0_{vi}} := v^0_c(c^0_{vi}, l_{0i}) = \lambda_{0i} \tag{9}$$

$$\frac{\delta V_i}{\delta c^1_{vi}} := v^1_c(c^1_{vi}, l_{1i}) = \lambda_{1i} \tag{10}$$

$$\frac{\delta V_i}{\delta l_{0i}} := R_1 v^0_l(c^0_{vi}, l_{0i}) + \lambda_{0i}(1-\tau)(\gamma_1 w_{01}) = 0 \tag{11}$$

$$\frac{\delta V_i}{\delta l_{1i}} := R_2 v^1_l(c^1_{vi}, l_{1i}) + \lambda_{1i}(1-\tau)(\gamma_2 w_{11}) = 0 \tag{12}$$

Return

Lawson (2017) Model

- Consumption Euler Equation

$$u_c'(c_{ci}) = v_c'(c_{vi}, l_{1i}) + \mu_i \tag{13}$$

- If the liquidity constrinat does not bind ($\mu_i = 0$ then MU_c today is equal to the MU when I enroll and smooth consumption with labour hours l_{1i}. The growth on MU depends only on the discount factor R.
- If liquidity constraints do bind ($\mu_i > 0$ which implies $a_{t+1} = a_t$) then $MU_c > v_c^1(c_{vi}^1, l_{1i})$: MU today is too high (in utils) μ_i measures how much I would pay to relax the liquidity constraint by 1\$.
- Distortion of human capital investment:
  - she may forego college not because its return is too low, but because she cannot finance consumption during enrollment

Return

Lawson (2017) Model

- Intertemporal labour-consumption conditions

$$w_{11} = -\frac{R_2}{(1\tau)\gamma_2}\frac{v_l^1(c_{vi}^1, l_{1i})}{v_c^1(c_{vi}^1, l_{1i})} \tag{14}$$

- we work until the value of extra labour hours equal the wage for those who went to college.
- because $w_{11} > w_{01}$ graduates supply labour at a higher marginal reward.

$$w_{01} = -\frac{R_1}{(1\tau)\gamma_1}\frac{v_l^0(c_{vi}^0, l_{0i})}{v_c^0(c_{vi}^0, l_{0i})} \tag{15}$$

- work until the value of extra hour of work of labour equals the no-grad wage

Return

Lawson (2017) Model

- The government chooses b and τ subject to BC:

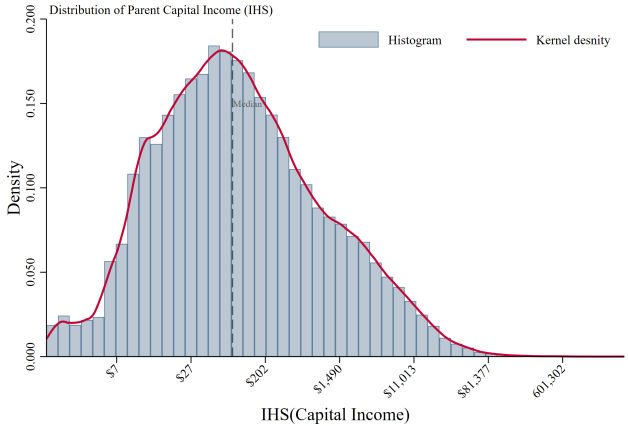
$$S(b+p)+G=\tau[S\gamma_2 E(Y_{11i}|s_i=1)+(1-S)\gamma_1 E(Y_{01i}|s_i=0)]=\tau Y \tag{16}$$

- To solve emperically-implementable version of

$$\frac{dW}{db}=\frac{\delta V}{\delta b}+\frac{\delta V}{\delta \tau}\frac{\delta \tau}{\delta b} \tag{17}$$

Return

Distribution of Capital Income



N = 100840. IHS(x) = ln(x+ √(x²+1)).
X-axis labels show approximate dollar equivalents.

Return